

Investigation of the Continuum Limit in the Presence of Small Breaking of Gauge Invariance in Lattice Gauge Theories

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This thesis investigates whether and how the continuum limit of a lattice gauge theory is affected by small explicit violations of gauge invariance. The study combines theoretical scaling arguments with numerical measurements of Wilson loops, Creutz ratios, the static potential, the Sommer scale, and related dimensionless observables. The central goal is to determine whether gauge-breaking effects vanish, persist, or modify the extrapolation toward the continuum limit as the lattice spacing is reduced.

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Notation and Conventions

a Lattice spacing.

β Bare lattice coupling parameter.

ε Parameter controlling the explicit breaking of gauge invariance.

$W(R, T)$ Wilson loop with spatial extent R and temporal extent T .

$\chi(R, T)$ Creutz ratio used to estimate the force and string tension.

r_0 Sommer scale defined through $r^2 F(r) = 1.65$.

σ String tension.

1. Introduction

1.1. Context and Problem

Gauge invariance is one of the defining structural principles of lattice gauge theory. In Wilson's formulation it is preserved exactly at nonzero lattice spacing, which strongly constrains the allowed lattice artifacts and supports a controlled approach to the continuum limit. In practical and generalized settings, however, gauge invariance may be only approximate. Small explicit violations can arise from regulator artifacts, effective descriptions, algorithmic imperfections, or imperfect implementations of gauge constraints.

The central problem of this thesis is therefore not whether exact gauge invariance is useful, but how robust the continuum limit is when the lattice theory is perturbed by a controlled, small gauge-breaking deformation. This requires a comparison between the standard gauge-invariant theory and deformed ensembles along a sequence of lattice spacings.

1.2. Research Gap

The theoretical expectation depends on the scaling character of the gauge-breaking deformation. If the deformation acts like an irrelevant lattice artifact, its effect should vanish as the lattice spacing is reduced. If it contains a relevant component, it may change the continuum theory unless it is tuned away. The numerical study must therefore distinguish a harmless change of cutoff effects from a genuine shift of continuum observables.

1.3. Aim and Research Questions

The aim of the thesis is to test this distinction numerically for pure $SU(2)$ lattice gauge theory with a small explicit gauge-breaking deformation. The guiding research question is:

Does a small explicit breaking of gauge invariance disappear in the continuum limit, or does it alter the continuum extrapolation of physical observables?

1.4. Strategy and Main Observables

The numerical strategy is to compare gauge-invariant and gauge-broken ensembles through observables that have a well-defined continuum interpretation. The main observables are temporal Wilson loops, Creutz ratios, the static potential $V(R)$, the Sommer scale r_0 , and the string tension σ or dimensionless combinations derived from it. These quantities provide both scale setting information and direct probes of the continuum extrapolation.

1.5. Structure of the Thesis

Chapter 2 summarizes the theoretical background needed for the analysis. Chapter 3 derives the concrete research gap and formulates the gauge-breaking framework and working hypotheses. Chapter 4 describes the simulation setup, observable extraction, continuum extrapolation, and uncertainty estimation. Chapter 5 presents the numerical results. Chapter 6 interprets the findings in relation to the research question and the theoretical expectations. The final chapter summarizes the conclusions and identifies directions for further research.

2. Scientific Background and Theoretical Framework

The aim of this chapter is to collect the theoretical ingredients needed for the analysis of the continuum limit in the presence of small explicit violations of gauge invariance. The discussion starts from continuum Yang–Mills theory and then introduces the lattice formulation, Wilson loops, the static potential, Creutz ratios, and the Sommer scale. These are the central concepts used later to define and compare physical observables at different lattice spacings.

2.1. Gauge Theories in the Continuum

2.1.1. Gauge Fields and Local Gauge Symmetry

A non-Abelian gauge theory is built from fields that transform locally under a Lie group. In this thesis the relevant case is the group $SU(2)$. A continuum gauge field can be written as

$$A_\mu(x) = A_\mu^a(x)T^a, \quad (2.1)$$

where T^a are the generators of the Lie algebra of $SU(2)$. A common choice is $T^a = \sigma^a/2$, with σ^a the Pauli matrices. The gauge field defines the covariant derivative

$$D_\mu = \partial_\mu + iA_\mu, \quad (2.2)$$

and the associated field-strength tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]. \quad (2.3)$$

The commutator term distinguishes a non-Abelian gauge theory from an Abelian one. It implies that the gauge field itself carries charge under the gauge group and therefore self-interacts.

A local gauge transformation is specified by a group element

$$\Omega(x) \in SU(2) \quad (2.4)$$

at every point in space-time. Under such a transformation, matter fields in the fundamental representation transform as

$$\psi(x) \mapsto \Omega(x)\psi(x), \quad (2.5)$$

while the gauge field transforms inhomogeneously,

$$A_\mu(x) \mapsto \Omega(x) A_\mu(x) \Omega^\dagger(x) - i \left(\partial_\mu \Omega(x) \right) \Omega^\dagger(x). \quad (2.6)$$

The inhomogeneous derivative term is necessary because the transformation is local. The field strength transforms covariantly,

$$F_{\mu\nu}(x) \mapsto \Omega(x) F_{\mu\nu}(x) \Omega^\dagger(x), \quad (2.7)$$

so traces of products of field strengths at the same point are gauge invariant.

In Euclidean space-time, the pure Yang–Mills action can be written as

$$S_{\text{YM}} = \frac{1}{2g^2} \int d^4x \operatorname{tr} F_{\mu\nu}(x) F_{\mu\nu}(x), \quad (2.8)$$

with summation over repeated Lorentz indices. This action is invariant under local gauge transformations. The coupling g controls the strength of the interaction. For an asymptotically free gauge theory, the effective coupling becomes small at short distances, while the long-distance theory is strongly coupled and requires non-perturbative methods.

2.1.2. Gauge-Invariant Observables

Physical observables in a gauge theory must be gauge invariant. Gauge-dependent quantities can be useful after a gauge fixing has been imposed, but they do not directly correspond to physical observables unless the gauge dependence is properly controlled. For pure gauge theory, important gauge-invariant observables include traces of closed parallel transporters, local action-density operators, correlation functions of gauge-invariant operators, and quantities derived from Wilson loops.

A central continuum object is the path-ordered parallel transporter along a curve C ,

$$U(C) = \mathcal{P} \exp \left(i \int_C dx^\mu A_\mu(x) \right). \quad (2.9)$$

For a closed curve, the trace

$$W(C) = \frac{1}{N_c} \operatorname{Re} \operatorname{tr} U(C) \quad (2.10)$$

is gauge invariant. On the lattice this object becomes the Wilson loop and is one of the most direct probes of confinement and of the static quark–antiquark potential.

2.1.3. Physical Consequences of Gauge Invariance

Gauge invariance is not only a formal redundancy of the description. It constrains the allowed terms in the action, the form of correlation functions, and the possible continuum limits. In the lattice formulation, preserving gauge invariance exactly at nonzero lattice spacing is usually advantageous because it prevents gauge-noninvariant counterterms

from being generated by radiative corrections. Wilson’s lattice gauge action realizes this idea by putting compact group-valued variables on links and by constructing the action from traces of closed loops [1, 2, 3].

The purpose of this thesis is to study what happens when this exact symmetry is slightly violated by an explicit deformation. From a renormalization-group point of view, the key question is whether the deformation behaves like an irrelevant lattice artifact, so that its effects vanish as $a \rightarrow 0$, or whether it defines a relevant perturbation that changes the continuum limit. This question is not purely formal: small violations of gauge constraints can appear as regulator artifacts, algorithmic imperfections, or implementation errors in effective or quantum-simulation formulations of lattice gauge theories [4, 5].

2.2. Lattice Gauge Theory

2.2.1. Discretization of Space-Time

In lattice gauge theory, Euclidean space-time is replaced by a hypercubic lattice

$$\Lambda_a = \{x = an \mid n \in \mathbb{Z}^4\}, \quad (2.11)$$

where a is the lattice spacing. The lattice spacing acts as an ultraviolet cutoff. Momenta larger than order $1/a$ are not resolved. The continuum limit is obtained by sending

$$a \rightarrow 0 \quad (2.12)$$

while keeping physical quantities fixed.

In a numerical simulation the lattice has finite extent

$$L_\mu = N_\mu a, \quad (2.13)$$

where N_μ is the number of lattice points in direction μ . Therefore two limits are conceptually distinct: the continuum limit $a \rightarrow 0$ and the infinite-volume limit $L_\mu \rightarrow \infty$. In practice, one tries to choose volumes large enough that finite-volume effects are smaller than the statistical and systematic errors relevant for the analysis. In the simulation campaign of this thesis, spatial volumes are organized to keep the physical spatial extent approximately fixed where possible, while the temporal extent is chosen as part of the ensemble definition and analysis strategy. [Check against the actual simulation setup: in the simulations in this thesis I use $L_\mu = \text{const}$ for the spatial directions and $N_\mu = \text{const}$ for the time direction.]

2.2.2. Link Variables and Plaquettes

The fundamental degrees of freedom of Wilson lattice gauge theory are link variables

$$U_\mu(x) \in \text{SU}(2), \quad (2.14)$$

which live on the oriented link from x to $x + a\hat{\mu}$. They are the lattice analogues of parallel transporters. For a smooth continuum field one may formally write

$$U_\mu(x) \approx \exp \left[iaA_\mu \left(x + \frac{a}{2}\hat{\mu} \right) \right]. \quad (2.15)$$

A local lattice gauge transformation assigns a group element $\Omega(x)$ to each site and acts on the link variables as

$$U_\mu(x) \mapsto \Omega(x)U_\mu(x)\Omega^\dagger(x + a\hat{\mu}). \quad (2.16)$$

The elementary closed loop on the lattice is the plaquette,

$$U_{\mu\nu}(x) = U_\mu(x)U_\nu(x + a\hat{\mu})U_\mu^\dagger(x + a\hat{\nu})U_\nu^\dagger(x). \quad (2.17)$$

Under a gauge transformation, the plaquette transforms covariantly at its base point,

$$U_{\mu\nu}(x) \mapsto \Omega(x)U_{\mu\nu}(x)\Omega^\dagger(x). \quad (2.18)$$

Therefore $\text{Re tr } U_{\mu\nu}(x)$ is gauge invariant.

For smooth fields, the plaquette has the continuum expansion

$$U_{\mu\nu}(x) = \exp \left[ia^2 F_{\mu\nu}(x) + \mathcal{O}(a^3) \right]. \quad (2.19)$$

Thus the plaquette is the lattice object from which the continuum field strength is recovered. Although the later analysis focuses on Wilson loops, static potentials, and Creutz ratios, this relation explains why the Wilson plaquette action reproduces Yang–Mills theory in the continuum limit.

2.2.3. Wilson Gauge Action

The standard Wilson gauge action for $\text{SU}(N_c)$ is

$$S_W[U] = \beta \sum_x \sum_{\mu < \nu} \left(1 - \frac{1}{N_c} \text{Re tr } U_{\mu\nu}(x) \right), \quad (2.20)$$

with

$$\beta = \frac{2N_c}{g_0^2}. \quad (2.21)$$

For $\text{SU}(2)$ this becomes

$$\beta = \frac{4}{g_0^2}, \quad (2.22)$$

and

$$S_W[U] = \beta \sum_x \sum_{\mu < \nu} \left(1 - \frac{1}{2} \text{Re tr } U_{\mu\nu}(x) \right). \quad (2.23)$$

The action is built only from traces of closed loops and is therefore exactly gauge invariant at nonzero lattice spacing.

The continuum limit of the Wilson action is obtained by inserting the plaquette expansion. Up to an irrelevant additive constant and lattice artifacts,

$$S_W[U] \longrightarrow \frac{1}{2g_0^2} \int d^4x \operatorname{tr} F_{\mu\nu}(x) F_{\mu\nu}(x). \quad (2.24)$$

This reproduces the Euclidean Yang–Mills action. The corrections at finite lattice spacing are cutoff effects. For the pure Wilson gauge action, the leading gauge-invariant lattice artifacts in many on-shell quantities are of order a^2 , although the actual scaling observed in numerical data can be affected by statistics, finite-volume effects, operator definitions, and the accessible range of lattice spacings.

2.2.4. Path Integral and Expectation Values

The lattice path integral is finite-dimensional at finite lattice spacing and finite volume. The partition function is

$$Z = \int \prod_{x,\mu} dU_\mu(x) e^{-S[U]}, \quad (2.25)$$

where $dU_\mu(x)$ is the Haar measure on $SU(2)$. The expectation value of an observable $\mathcal{O}[U]$ is

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \prod_{x,\mu} dU_\mu(x) \mathcal{O}[U] e^{-S[U]}. \quad (2.26)$$

Gauge-invariant expectation values can be computed without gauge fixing. This is one of the practical advantages of the compact Wilson formulation.

In Monte Carlo simulations the path integral is estimated by generating an ensemble of gauge-field configurations distributed according to $e^{-S[U]}$. For an ensemble of N_{cfg} configurations U_i , the estimator is

$$\langle \mathcal{O} \rangle \approx \frac{1}{N_{\text{cfg}}} \sum_{i=1}^{N_{\text{cfg}}} \mathcal{O}[U_i]. \quad (2.27)$$

Because successive configurations in a Markov chain are correlated, autocorrelations must be considered when estimating uncertainties.

2.2.5. Continuum Limit and Renormalization

The continuum limit is not obtained by simply decreasing a at fixed bare coupling. Instead, the bare parameters have to be tuned so that physical observables remain fixed in physical units. For pure $SU(2)$ gauge theory with the Wilson action, decreasing the lattice spacing corresponds to increasing β , or equivalently decreasing the bare coupling g_0 .

A useful way to formulate the continuum limit is in terms of a physical length scale, such as the Sommer scale r_0 . The dimensionless ratio

$$\frac{r_0}{a} \quad (2.28)$$

is measured on the lattice. Increasing r_0/a means that the same physical length is resolved by more lattice points, i.e. the lattice spacing is smaller in physical units. Therefore continuum extrapolations can be expressed as fits in

$$\left(\frac{a}{r_0}\right)^2 \quad (2.29)$$

or another measured dimensionless lattice-spacing proxy.

Continuum extrapolations compare observables at fixed physical conditions, not merely at fixed lattice indices. For observables derived from Wilson loops this means that a loop size R on a coarse ensemble and the same integer R on a fine ensemble do not represent the same physical distance. Therefore one should compare at fixed ratios such as r/r_0 or other fixed dimensionless distances. This point is emphasized in recent work on controlled continuum extrapolations of Wilson-loop observables, where Creutz ratios are used as finite continuum quantities and fitted at fixed physical distance [6].

The renormalization-group interpretation is central for the present thesis. Different lattice actions can lead to the same continuum theory if they differ only by irrelevant operators. In that case they lie in the same universality class. By contrast, a relevant deformation changes the long-distance theory unless its coefficient is tuned to zero. A gauge-breaking term must therefore be classified by its effect on long-distance, gauge-invariant observables as $a \rightarrow 0$. The numerical analysis in later chapters tests this question directly.

2.3. Static Potential and Confinement

2.3.1. Wilson Loops

A Wilson loop is the trace of the ordered product of link variables around a closed contour. For a rectangular loop in the $\mu\nu$ -plane with side lengths L_μ and L_ν in lattice units, the code measures the normalized real trace

$$W_{\mu\nu}(L_\mu, L_\nu) = \frac{1}{N_c V} \sum_x \text{Re tr} \left[U_\mu(x; L_\mu) U_\nu(x + L_\mu \hat{\mu}; L_\nu) U_\mu^\dagger(x + L_\nu \hat{\nu}; L_\mu) U_\nu^\dagger(x; L_\nu) \right], \quad (2.30)$$

where V is the lattice volume and

$$U_\mu(x; L_\mu) = U_\mu(x) U_\mu(x + \hat{\mu}) \cdots U_\mu(x + (L_\mu - 1)\hat{\mu}) \quad (2.31)$$

denotes a straight Wilson line of length L_μ in direction μ . The factor $1/N_c$ normalizes the trace, while the factor $1/V$ averages over all possible base points of the loop.

For temporal Wilson loops, the last lattice direction is used as the Euclidean time direction,

$$t = N_d - 1. \quad (2.32)$$

The code computes rectangular loops in each spatial-time plane and averages over the spatial directions:

$$W(R, T) = \frac{1}{N_d - 1} \sum_{\mu=0}^{N_d-2} W_{\mu,t}(R, T). \quad (2.33)$$

For the SU(2) simulations considered in this thesis, $N_c = 2$, so the stored temporal Wilson loop is

$$W(R, T) = \frac{1}{2V(N_d - 1)} \sum_{\mu=0}^{N_d-2} \sum_x \text{Re tr } W_{\mu,t}(x; R, T). \quad (2.34)$$

Thus the values written to the output file are already normalized by the color factor, averaged over the full lattice volume, and averaged over spatial directions.

In the implementation, the rectangular loop is constructed from two paths that connect opposite corners of the rectangle. The first path is

$$l_\mu = U_\mu(x; L_\mu) U_\nu(x + L_\mu \hat{\mu}; L_\nu), \quad (2.35)$$

and the second path is

$$l_\nu = U_\nu(x; L_\nu) U_\mu(x + L_\nu \hat{\nu}; L_\mu). \quad (2.36)$$

The measured loop contribution is then

$$\text{Re tr } (l_\mu l_\nu^\dagger). \quad (2.37)$$

This is equivalent to the trace of the ordered product around the rectangular closed contour.

The temporal Wilson-loop output file has the structure

```
# step, L, T, W_temp
```

where L is the spatial extent R , T is the temporal extent, and W_temp is the averaged, normalized value defined in Eq. (2.33).

2.3.2. Effective Potential Extraction

Temporal Wilson loops are used to extract the static quark–antiquark potential. For large Euclidean time extent T , the Wilson loop has the spectral form

$$W(R, T) = \sum_n c_n(R) e^{-aE_n(R)T}, \quad (2.38)$$

where $E_0(R) = V(R)$ is the static ground-state potential and the higher $E_n(R)$ describe excited states with the same static sources. At sufficiently large T , the ground state dominates:

$$W(R, T) \propto e^{-aV(R)T}. \quad (2.39)$$

A lattice effective potential is therefore defined by the ratio

$$aV_{\text{eff}}(R, T) = -\ln \left[\frac{W(R, T+1)}{W(R, T)} \right]. \quad (2.40)$$

If excited-state contributions are negligible, this approaches the desired static potential in lattice units:

$$aV_{\text{eff}}(R, T) \xrightarrow{T \rightarrow \infty} aV(R). \quad (2.41)$$

In the numerical analysis, $W(R, T)$ denotes the already normalized and spatially averaged observable stored as `W_temp`. No additional color, volume, or spatial-direction normalization is applied during the analysis. The main remaining analysis choices are the thermalization cut, the treatment of autocorrelations, the temporal range used for the extraction, and the estimation of statistical uncertainties.

2.3.3. Creutz Ratios

Creutz ratios are combinations of neighboring Wilson loops designed to cancel some perimeter and normalization contributions. In the continuum they can be viewed as the mixed derivative

$$\chi(r, t) = -\frac{\partial^2}{\partial r \partial t} \ln W(r, t), \quad (2.42)$$

where $W(r, t)$ is a rectangular Wilson loop of physical spatial extent r and Euclidean time extent t . A central finite-difference discretization gives a lattice Creutz ratio located at half-integer separations,

$$\chi\left(R + \frac{1}{2}, T + \frac{1}{2}\right) = \ln \left[\frac{W(R+1, T)W(R, T+1)}{W(R, T)W(R+1, T+1)} \right], \quad (2.43)$$

or, equivalently after a shift of indices,

$$\chi(R, T) = -\ln \left[\frac{W(R, T)W(R-1, T-1)}{W(R, T-1)W(R-1, T)} \right]. \quad (2.44)$$

The normalization of the Wilson loop, including the factor $1/N_c$, cancels in this ratio. This is useful for the present SU(2) analysis because the same formula can be applied directly to the normalized Wilson-loop output described above.

The central-difference form approximates the continuum mixed derivative with cutoff effects of order a^2 when the underlying Wilson loop is smooth. This makes Creutz ratios particularly useful for continuum studies: they are finite observables, their leading normalization and perimeter contributions cancel, and they are closely related to the static force. In the large-time limit,

$$\frac{\chi(R, T)}{a^2} \xrightarrow{T \rightarrow \infty} F(r) \quad (2.45)$$

up to the usual distinction between the discrete lattice midpoint and the physical distance r . Diagonal ratios $\chi(R, R)$ are also useful summary observables because they probe the same spatial and temporal scale and are straightforward to compare across ensembles [7, 6].

If large Wilson loops obey an area law,

$$W(R, T) \sim \exp[-\sigma a^2 RT - \text{perimeter terms}], \quad (2.46)$$

then the Creutz ratio approaches

$$\chi(R, T) \xrightarrow{R, T \rightarrow \infty} \sigma a^2. \quad (2.47)$$

Thus Creutz ratios provide a direct lattice estimator for the string tension in lattice units.

At finite R and T , Creutz ratios contain corrections from excited states, Coulombic contributions, finite-volume effects, and lattice artifacts. They are therefore best interpreted either as approximate string-tension estimators at large loop sizes or as diagnostic observables for scaling comparisons between ensembles.

For continuum extrapolations the important distinction is between fixed lattice separation and fixed physical distance. Holding R fixed while taking $a \rightarrow 0$ drives the physical distance $r = Ra$ to zero and therefore probes the short-distance limit rather than the same physical force. The continuum comparison should instead use a dimensionless distance such as

$$\hat{r} = \frac{r}{r_0} \quad (2.48)$$

and interpolate the measured Creutz ratios to common $\hat{r}$ values before fitting in a/r_0 . This is also where gauge-breaking effects are expected to be most visible: short-distance Wilson-loop observables are sensitive to ultraviolet deformations, while large loops suffer from rapidly increasing statistical noise.

2.3.4. Sommer Scale and String Tension

The static potential can also be used to set the scale. A common phenomenological form is the Cornell-type potential

$$V(r) = V_0 - \frac{\alpha}{r} + \sigma r, \quad (2.49)$$

where V_0 is an additive self-energy, α parametrizes the short-distance Coulombic part, and σ is the string tension. The corresponding force is

$$F(r) = \frac{dV}{dr}. \quad (2.50)$$

The additive constant V_0 drops out of the force.

The Sommer scale r_0 is defined implicitly by

$$r_0^2 F(r_0) = 1.65. \quad (2.51)$$

This definition uses the force at intermediate distances, where the potential can often be determined more accurately than at very large distances. In lattice units, the force is estimated from finite differences of the potential. If $\hat{V}(R) = aV(Ra)$, then

$$\hat{F}\left(R + \frac{1}{2}\right) = \hat{V}(R+1) - \hat{V}(R) \quad (2.52)$$

is a dimensionless estimate of $a^2 F(r)$ at an intermediate distance. The Sommer condition can then be written as

$$\left(\frac{r}{a}\right)^2 \hat{F}\left(\frac{r}{a}\right) = 1.65. \quad (2.53)$$

Solving this equation gives r_0/a .

The ratio r_0/a is one of the main scale-setting quantities in this thesis. It allows data at different β values to be compared in approximately physical units and provides the continuum variable

$$\frac{a}{r_0} = \left(\frac{r_0}{a}\right)^{-1}. \quad (2.54)$$

The continuum limit corresponds to

$$\frac{a}{r_0} \rightarrow 0. \quad (2.55)$$

The string tension may also be expressed in dimensionless form as

$$\sigma r_0^2. \quad (2.56)$$

This quantity is useful because it is independent of the arbitrary choice of lattice units. Agreement of such dimensionless quantities along a sequence of lattice spacings is a central test of scaling toward a common continuum limit.

3. Research Gap and Gauge-Breaking Framework

3.1. Research Gap Derived from the Background

The background chapter establishes the theoretical role of exact lattice gauge invariance and the observables used to monitor the continuum limit. The open question for the present work is how a controlled violation of this symmetry enters the scaling analysis. The deformation must be specified at the level of the lattice action or update rule, and its effect must then be tested through dimensionless observables that can be compared across lattice spacings.

3.2. Definition of the Gauge-Breaking Deformation

Define the modified lattice action or update rule used in the simulations. Specify the breaking parameters, for example ε_1 and ε_2 , and explain which gauge transformations are no longer exact symmetries.

3.3. Symmetry and Scaling Expectations

3.3.1. Relevant, Marginal, and Irrelevant Breaking Terms

Classify the deformation in renormalization-group language and state which scaling behavior would be expected for each case.

3.3.2. Possible Restoration of Gauge Invariance

Explain under which conditions gauge-invariant observables can approach the same continuum limit even when gauge invariance is broken at finite lattice spacing.

3.3.3. Expected Dependence on Breaking Strength and Lattice Spacing

State the expected dependence on ε_1 , ε_2 , and a , and use this expectation to motivate the fit forms introduced in the methodology chapter.

3.4. Working Hypotheses

The numerical analysis tests the following hypotheses:

1. Gauge-invariant observables approach the same continuum values for sufficiently small ε .
2. Gauge-breaking effects scale to zero with a measurable power of a or ε .
3. Above a threshold in ε , continuum extrapolations become unstable or incompatible with the gauge-invariant limit.

4. Methodological Approach

4.1. Simulation Setup and Parameters

The gauge-field ensembles were generated for pure SU(2) lattice gauge theory on four-dimensional hypercubic lattices. Each ensemble is specified by an `input.yaml` file, which contains the lattice geometry $N_0 \times N_1 \times N_2 \times N_3$, the bare coupling β , the gauge-breaking parameters ε_1 and ε_2 , the Metropolis update parameters, and the measurement setup. The simulations use the Wilson gauge action supplemented by the gauge-breaking deformation introduced in Chapter 3. Gauge configurations are generated with a local Metropolis algorithm. One sweep consists of updating all link variables of the lattice, with `nHits` Metropolis proposals per link. The proposal size is controlled by the parameter `delta`. The main bare parameters varied in the simulation campaign are the coupling β , which controls the lattice spacing, and the breaking strength ε_1 . The parameter ε_2 is kept at zero unless stated otherwise. Measurements are performed after fixed sweep intervals and include the plaquette, the average link trace, and temporal Wilson loops. The temporal Wilson loops are measured for selected spatial and temporal extents (R, T) and are written to `W_temp.out`; these data form the basis for the extraction of the static potential, Creutz ratios, and scale setting quantities.

4.2. Ensemble Organization and Fixed-Physics Logic

Explain how ensembles are grouped by common physical and algorithmic parameters. State which parameters are varied in the continuum study and which are held fixed. The organizing principle is that comparisons across β should be made at fixed physical conditions as far as the available data allow, rather than at fixed lattice indices alone.

4.3. Thermalization, Autocorrelations, and Bootstrap

4.3.1. Thermalization Cuts

Describe how the initial non-equilibrated part of each Markov chain is identified and removed before combining runs.

4.3.2. Integrated Autocorrelation Time

Estimate or monitor autocorrelation times for representative observables in order to choose measurement spacing and bootstrap block lengths.

4.3.3. Bootstrap Blocking

The measurements after thermalization are still not statistically independent, because consecutive configurations are generated by a Markov chain. A naive bootstrap over individual measurements would therefore underestimate the statistical error. To account for this, the ordered measurement history is divided into contiguous blocks of fixed length b in simulation time. The bootstrap samples are then generated by drawing these blocks with replacement and averaging over the resampled history. Unless stated otherwise, the statistical errors quoted in this work use $N_{\text{boot}} = 1000$ bootstrap replicas.

The block length must be large compared with the autocorrelation time of the observables used in the analysis. If b is too small, correlations remain between the resampled data and the error estimate is biased downward. If b is too large, the number of available blocks becomes small and the bootstrap estimate becomes noisy. In practice, several block sizes are compared and the chosen value is taken from the region where the estimated errors are stable under further increases of b .

The same bootstrap samples are propagated through the nonlinear analysis steps. Wilson loops are resampled first, and derived quantities such as effective potentials, the static potential $V(R)$, Creutz ratios, and the Sommer scale are recomputed for each bootstrap replica. This preserves correlations between quantities derived from the same ensemble and gives the statistical uncertainties used in the later continuum analysis.

4.4. Analysis Pipeline and Data Reduction

The final analysis starts from the individual simulation runs and groups together runs with the same base parameters, in particular the same β , lattice size, ε_1 , and ε_2 . Runs in such a group differ only by statistical details such as the random seed or the total number of sweeps and are treated as independent Markov chains for the same ensemble.

For each input run the early Monte Carlo history is inspected before combining the data. In practice, Wilson loops and simple observables are plotted as a function of the simulation step. A thermalization cutoff is chosen from these plots. Measurements before this cutoff are discarded, and the remaining measurements from all runs in the group are combined into one data set.

The combined post-thermalization data are then used to estimate the Wilson loops $W(R, T)$ and their statistical uncertainties. Errors are obtained by a blocked bootstrap, with a block size chosen large enough to reduce the effect of autocorrelations. From the Wilson loops, effective masses are formed and fit windows in T are selected to extract the static potential $V(R)$.

The finalized potential points are fitted with a Cornell-type ansatz to obtain the Sommer scale r_0/a . This scale is then used to form the dimensionless lattice spacing

$$\hat{a} = \frac{a}{r_0} \tag{4.1}$$

and the rescaled breaking parameter

$$\bar{\varepsilon} = \frac{\varepsilon_1}{\beta} \left(\frac{r_0}{a} \right)^2. \quad (4.2)$$

Creutz ratios and available gradient-flow observables are evaluated from the same combined ensemble. The resulting finalized ensemble summaries are the inputs for the later comparison across different β values and breaking strengths.

4.5. Extraction of Physical Observables

4.5.1. Wilson-Loop Analysis

Describe the effective-mass plateaus, the choice of temporal fit windows, and the systematic uncertainty associated with varying those windows.

4.5.2. Static Potential

Describe the potential extraction, Cornell-type fits, force determination, and Sommer-scale extraction.

4.5.3. Creutz-Ratio Analysis

Define the Creutz-ratio observables used in the numerical analysis, explain their large-time interpretation, and state how they are related to the string tension.

4.5.4. Scale Setting

Summarize the determination of the Sommer scale in lattice units, possible alternative scale estimates, and consistency checks.

4.6. Continuum Extrapolation and Scaling Tests

4.6.1. Choice of Continuum Variable

The primary continuum variable is chosen to be the measured dimensionless lattice spacing

$$\hat{a} = \frac{a}{r_0}. \quad (4.3)$$

For observables that are expected to have leading even-power cutoff effects with the Wilson gauge action, the default horizontal axis is therefore $\hat{a}^2 = (a/r_0)^2$. Using the measured scale rather than the bare coupling makes the extrapolation closer to a fixed-physics comparison and reduces the risk of interpreting changes in scale setting as physical changes in the observable itself.

For Creutz-ratio observables the quantity to extrapolate should be dimensionless. Convenient choices are

$$\hat{\chi}(\hat{r}) = r_0^2 \frac{\chi(r, t)}{a^2}, \quad \hat{r} = \frac{r}{r_0}, \quad (4.4)$$

or, for large loops where the string-tension interpretation is appropriate, σr_0^2 . The measured lattice ratio $\chi(R, T)$ is dimensionless and approaches σa^2 in the area-law regime; division by a^2 and multiplication by r_0^2 converts it to a quantity with a finite physical continuum limit. The same logic applies to the force extracted from the static potential.

4.6.2. Fixed-Physics Trajectories

Ensembles at different β values are compared along trajectories where the physical volume and the dimensionless analysis point are kept as fixed as the available data allow. In practice this means comparing observables at common $\hat{r} = r/r_0$ and using only ensembles whose volumes are large enough that finite-volume corrections are not expected to dominate the Wilson-loop signal. When exact matching is not possible, the mismatch is included as a systematic effect or avoided by restricting the fit range.

The gauge-breaking parameters are treated as additional bare deformation parameters. If they are held fixed while a changes, the continuum fit tests whether their effect vanishes dynamically or survives in the continuum intercept. If a theoretical scaling prescription for ε is used, for example $\varepsilon \propto a^q$, the fit must instead include this scaling explicitly. The central diagnostic is whether the broken and unbroken data approach a common continuum value.

4.6.3. Fit Ansatz

The fit strategy compares a gauge-invariant baseline with combined lattice-spacing and breaking-dependence models.

Gauge-invariant baseline. For the gauge-invariant ensembles, a minimal continuum ansatz for a dimensionless observable $\hat{O}$ is

$$\hat{O}(\hat{a}) = O_0 + c_a \hat{a}^2 + c_{a4} \hat{a}^4 + \dots \quad (4.5)$$

The $\hat{a}^4$ term is included only when the data support it or when it is used as a fit-systematic variation. The default fit should be stable under removing the coarsest lattice spacing and under modest changes of the physical-distance range used for the observable.

Combined lattice-spacing and breaking dependence. A useful shared-intercept test is

$$\hat{O}(\hat{a}, \varepsilon) = O_0 + c_a \hat{a}^2 + c_\varepsilon \varepsilon \hat{a}^p + c_{a\varepsilon} \varepsilon \hat{a}^{p+2} + \dots, \quad (4.6)$$

with $p > 0$. This ansatz encodes the hypothesis that gauge breaking is an irrelevant cutoff effect: for any fixed small ε the breaking terms vanish as $\hat{a} \rightarrow 0$, and all data share the same continuum intercept O_0 .

The competing separate-intercept test is

$$\hat{O}(\hat{a}, \varepsilon) = O_0 + c_\varepsilon \varepsilon + c_a \hat{a}^2 + c_{a\varepsilon} \varepsilon \hat{a}^2 + \dots \quad (4.7)$$

Here a statistically significant c_ε indicates that the breaking deformation changes the continuum value of the observable rather than merely changing cutoff effects. If the deformation can enter symmetrically in the sign of ε , a quadratic term $c_{\varepsilon^2} \varepsilon^2$ should be included as a systematic variant.

This structure follows the same logic used in controlled continuum extrapolations with smearing or flow: one separates changes in the path toward the continuum from changes in the continuum observable itself [6]. In that setting, fixed smearing in lattice units can share the same continuum intercept if it remains in the controlled expansion regime, while a fixed physical smoothing radius defines a different continuum observable. In the present gauge-breaking problem there is no guarantee of a shared intercept, so Eqs. (4.6) and (4.7) are both part of the analysis.

4.6.4. Uncertainty Budget

Statistical errors. Statistical errors are obtained from blocked bootstrap samples. The block size is chosen large enough to reduce residual autocorrelation effects in the saved Markov-chain history. Derived observables such as Creutz ratios, Sommer scales, and continuum-fit parameters are recomputed inside the bootstrap whenever possible, so nonlinear error propagation is captured by the resampling.

Fit-window and interpolation systematics. Wilson-loop observables depend on analysis choices: the temporal range used for effective-potential plateaus, the loop sizes included in Creutz-ratio summaries, and the interpolation used to compare fixed physical distances. These choices are varied to estimate systematic effects. For Creutz ratios, short distances are most vulnerable to cutoff and breaking artifacts, while long distances are most vulnerable to statistical noise.

Finite-volume and continuum-fit systematics. Finite-volume effects are monitored by comparing ensembles with similar $\hat{a}$ and ε but different physical volumes when available. Continuum-fit systematics are estimated by changing the fit ansatz, removing the coarsest lattice spacing, and checking whether the extrapolation remains monotonic over the fitted range. Non-monotonic behavior or strong shifts of the intercept under reasonable fit variations are treated as evidence that the available lattice spacings are not yet in a controlled scaling regime.

4.7. Computational Environment

The simulations were carried out on the QBig cluster. Most production runs were performed on nodes 13–17 equipped with NVIDIA P100 GPUs. Additional and faster runs were performed on nodes 18 and 19 equipped with NVIDIA A100 GPUs. The GPU type affects the wall-clock runtime but not the generated Markov-chain ensemble, provided that the same action, random seed handling, and update algorithm are used.

5. Results

5.1. Thermalization and Data Quality

Insert representative thermalization histories and autocorrelation estimates.

5.2. Gauge-Invariant Baseline

Present the $\varepsilon = 0$ results for $W(R, T)$, $V(R)$, r_0 , and σ . This establishes the reference continuum behavior.

5.3. Dependence on Gauge-Breaking Strength

5.3.1. Observable Shifts at Fixed Coupling

Report how each observable changes with ε_1 at fixed β and fixed lattice geometry.

5.3.2. Scaling of Shifts with Breaking Strength

Fit or summarize whether the shifts are linear, quadratic, or nonlinear in the breaking strength within the accessible range.

5.3.3. Thresholds or Nonlinear Behavior

Identify any breaking strengths for which the data no longer follow the small-deformation pattern assumed in the scaling analysis.

5.4. Continuum Limit at Fixed Breaking

5.4.1. Continuum Extrapolations for Each Breaking Strength

Present continuum fits separately for each investigated breaking strength.

5.4.2. Comparison with the Gauge-Invariant Limit

Compare the extrapolated values with the $\varepsilon = 0$ baseline and state whether the intercepts are statistically compatible.

5.4.3. Stability Under Fit Variations

Show how the continuum conclusions change when fit ranges, lattice spacings, and systematic variants are modified.

5.5. Combined Scaling Analysis

Fit the dependence on both lattice spacing and gauge-breaking parameter. State whether the data support restoration of gauge invariance in the continuum limit.

5.6. Finite-Volume and Systematic Checks

Collect checks that are not part of the primary result but constrain the interpretation.

6. Discussion

6.1. Interpretation of the Continuum Limit

Discuss whether small gauge breaking behaves like an irrelevant lattice artifact, a relevant deformation, or a source of uncontrolled continuum bias.

6.2. Comparison with Theoretical Expectations

Relate the numerical findings to the hypotheses from Chapter [3](#).

6.3. Limitations

Discuss limits from statistics, accessible lattice spacings, finite volume, fit-systematic control, and the specific form of gauge breaking studied.

6.4. Implications

Explain what the result means for simulations or formulations where gauge invariance is only approximate.

7. Conclusion and Outlook

7.1. Summary of Findings

State the central quantitative conclusions.

7.2. Answer to the Research Question

Give the direct answer to whether the continuum limit is robust under the investigated small breaking of gauge invariance.

7.3. Future Work

Possible extensions:

- Study additional forms of gauge-breaking terms.
- Increase statistics and extend to finer lattice spacings.
- Add gauge-variant diagnostics to directly measure symmetry restoration.
- Compare different gauge groups or dimensions.

A. Simulation Tables

Provide complete tables of ensembles, including β , lattice size, breaking parameters, number of sweeps, thermalization cuts, and measured scales.

B. Additional Plots

Include supplementary Wilson-loop plateaus, bootstrap block-size scans, potential fits, and continuum-fit variants.

C. Analysis Code and Reproducibility

Document the exact command-line workflow, cache version, software environment, and data-selection rules used for the final analysis.

D. Derivations

Collect longer derivations that would interrupt the main text, such as Creutz-ratio identities or continuum-scaling arguments.

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